Basic Auton: Drive Team Guide

Basic Auton scores **LEAVE (3)** + **PARK (5)** in the 30-second AUTO period. If **both** robots on our alliance do it, that's 16 points and the **SWARM ranking point**. Route pictures: `docs/game/diagrams/basic-auton-near.pdf` and `basic-auton-far.pdf`.

1. Pick the right OpMode

On the Driver Station, open the **Basic Auton** group. Pick our alliance color and starting spot. **Every start is the same: the robot's BACK touches the wall.**

OpMode	Set the robot up...	Pick it when...
Basic Auton RED NEAR	Tile A6, BACK against the RED wall, facing into the field	We're red, and our partner starts somewhere else
Basic Auton RED FAR	Tile B1, BACK against the AUDIENCE wall, facing the far wall	We're red, and our partner wants the NEAR spot
Basic Auton BLUE NEAR	Tile F1, BACK against the BLUE wall, facing into the field	We're blue, and our partner starts somewhere else
Basic Auton BLUE FAR	Tile E6, BACK against the FAR wall, facing the audience	We're blue, and our partner wants the NEAR spot

Decide this with our partner before the match, using the **alliance planning sheet**. After AUTO, the matching **Main TeleOp** is already picked for you.

2. Press INIT: what the screen means

INIT gets the robot ready. **Nothing moves during INIT** (game rule G304.H): the robot must stay still until the match starts. You'll see something like this:

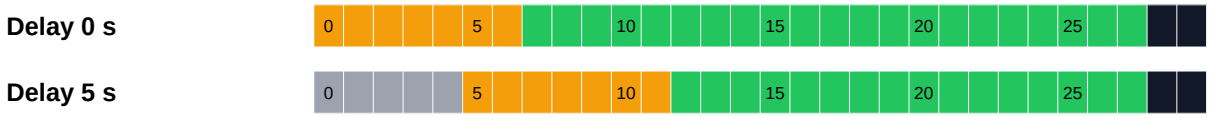
```
BASIC AUTON : RED  NEAR  (lights are RED)
SET UP: tile A6, BACK touching the RED wall, FRONT facing into the field
Wrong one? Stop and pick the right Basic Auton.
Start delay : 3 s   (dpad up/down to change)
Start position : x 60.0  y 63.0  heading -90°
Pinpoint : READY  (keep the robot STILL until READY)
Battery : 12.9 V
```

Line	What to check
BASIC AUTON + the **lights**	Both lights show our alliance color (red or blue). Wrong color or wrong NEAR/FAR? Press STOP and pick the right OpMode.
SET UP	Exactly how to place the robot. Check the robot matches it: right tile, BACK on the wall.
Start delay	How long the robot waits after START before it moves. See section 3.
Pinpoint	Must say **READY** before START. It's measuring where "still" is, so **don't touch or bump the robot** during INIT.
Battery	Low battery = slower, less accurate driving. Swap it if it's low.

3. The START DELAY: what it is and why we use it

The **start delay** makes our robot **wait** a few seconds after the match starts, and **then** drive its route. AUTO is 30 seconds, and our route only takes a few seconds, so we have time to spare.

Why wait? Both robots on our alliance drive at the same time, and they both park in the **same small LOADING ZONE** (about 23 in wide). If our partner's route crosses ours, the robots can **crash** and nobody parks. Waiting lets our partner go first, then we follow.



Grey = waiting (delay). Orange = driving the route (a few seconds; we'll time it in testing). Green = parked. Black = the code stops everything by 28 s, before AUTO ends at 30 s.

How to set it (during INIT, before START):

Do this	Result
Press **dpad UP** on gamepad 1	Delay goes up 1 second (max 10 s)
Press **dpad DOWN**	Delay goes down 1 second (min 0 s)
Look at the screen	"Start delay" shows the number. Check it before START!

The delay goes back to **0** every time you press INIT. Set it again for each match.

How to choose the number:

• **0 s**: our partner's route doesn't cross ours (most matches).
• **A few seconds**: our partner drives through our path, or parks in the LOADING ZONE first. Ask them how long their AUTO takes, and wait until they're out of the way.
• Write the agreed delay on the alliance sheet ("Our start delay").

4. After START

First the robot checks the **Pinpoint** is READY (up to 3 s). If it isn't, the robot **does not drive** (YELLOW lights), because driving without knowing where it is would be worse. Then it waits for the delay and drives each waypoint, slowing down as it gets close and stopping on the spot (no guessing with timers).

Lights at the end	Means	Do this
GREEN	Parked where planned	Nothing. Nice!
YELLOW	Stopped early: a waypoint took too long (bumped, blocked, slipping), or the Pinpoint wasn't READY	Tell the coders. Write it in the test log.

5. Quick checklist

	Before every AUTO
[ ]	Right OpMode: alliance color + NEAR/FAR agreed with our partner
[ ]	Robot on the right tile, its BACK touching the wall (the SET UP line on the screen)
[ ]	Press INIT. Both lights show our alliance color
[ ]	Start delay set (dpad up/down) and checked on the screen
[ ]	Pinpoint says READY. Hands off the robot
[ ]	Battery charged